ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/232084710

Characterization of copper clusters through the use of density functional theory reactivity descriptors Characterization of copper clusters through the use of density functional th...

Article *in* The Journal of Chemical Physics · August 2002

DOI: 10.1063/1.1493178

CITATIONS
204

READS
3,404

2 authors:

Pablo Jaque
University of Chile
82 PUBLICATIONS **2,497** CITATIONS

SEE PROFILE

Alejandro Toro-Labbé
Pontificia Universidad Católica de Chile
304 PUBLICATIONS **10,579** CITATIONS

SEE PROFILE

All content following this page was uploaded by Pablo Jaque on 28 May 2014.

The user has requested enhancement of the downloaded file.

The Journal of Chemical Physics

Characterization of copper clusters through the use of density functional theory reactivity descriptors

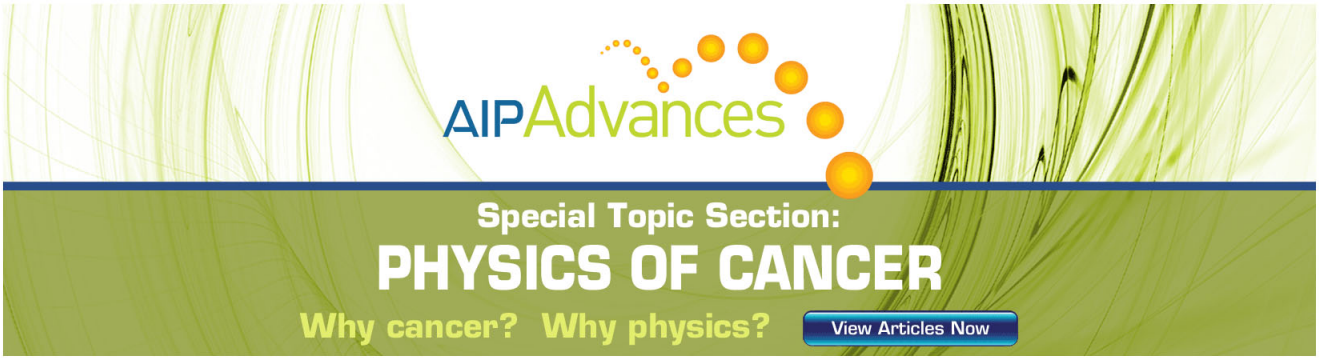
Pablo Jaque and Alejandro Toro-Labbé

Citation: J. Chem. Phys. **117**, 3208 (2002); doi: 10.1063/1.1493178
View online: http://dx.doi.org/10.1063/1.1493178
View Table of Contents: http://jcp.aip.org/resource/1/JCPSA6/v117/i7
Published by the American Institute of Physics.

Additional information on J. Chem. Phys.

Journal Homepage: http://jcp.aip.org/
Journal Information: http://jcp.aip.org/about/about_the_journal
Top downloads: http://jcp.aip.org/features/most_downloaded
Information for Authors: http://jcp.aip.org/authors

ADVERTISEMENT



Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

Characterization of copper clusters through the use of density functional theory reactivity descriptors

Pablo Jaque and Alejandro Toro-Labbé[a)]
Laboratorio de Química Teórica Computacional (QTC), Facultad de Química, Pontificia Universidad Católica de Chile, Casilla 306, Correo 22, Santiago, Chile

(Received 22 January 2002; accepted 21 May 2002)

In this paper we study nine neutral copper clusters through the theoretical characterization of their molecular structures, binding energy, electronic properties, and reactivity descriptors. Geometry optimization and vibrational analysis were performed using density functional theory calculations with a hybrid functional combined with effective core potentials. It is shown that reactivity descriptors combined with reactivity principles like the minimum polarizability and maximum hardness are operative for characterizing and rationalizing the electronic properties of copper clusters. © *2002 American Institute of Physics.* [DOI: 10.1063/1.1493178]

I. INTRODUCTION

The interest in the study of small metal clusters has grown considerably in the last years because new experimental and theoretical techniques have been developed allowing detailed characterizations of this type of systems.[1] Much information is now available concerning clusters' spectroscopy, structure, and their chemical reactivity toward small molecules. Transition metal clusters are particularly interesting for their potential use in many processes like heterogeneous catalysis, in organometallic chemistry or new electronic materials.[1–3]

Cluster properties are very sensitive to the number of atoms, and they can sometimes change dramatically with the addition or removal of one atom from the cluster. The research in cluster chemistry has been guided to study the dependence of different properties upon the size of the cluster, and hopefully how these properties tend to the bulk values. Properties such as ionization potentials, electron affinities, chemical reactivity, investigations of ion abundances, and dissociation energies of alkali-metal and transition-metal clusters have been reported.[4–9] In particular, copper clusters have been extensively studied both experimentally and theoretically.[10–29]

In this paper we provide a new viewpoint to understand and if possible to predict properties of atomic aggregates. In this context, the aim of this paper is to characterize neutral copper clusters in terms reactivity descriptors that have already been used with success in rationalizing different kinds of molecular structures and chemical reactions.[30,31] Density functional theory (DFT) (Ref. 32) has provided the basis for rigorous mathematical definitions of reactivity descriptors like chemical potential (μ), electronegativity (χ), chemical hardness (η), softness ($S=1/\eta$), etc. All of these are well established global quantities in chemical reactivity studies and are going to be used in this paper to characterize copper clusters. Although local and nonlocal quantities such as the condensed Fukui function,[32,33] local softness,[32] and electrophilicity[34] have also been introduced in the theory to characterize specific sites within a molecule, in the present study we are going to focus our attention on the study of global reactivity indexes.

The understanding of any molecular system and its reactivity requires the knowledge of the energy hypersurface, however, the computational cost involved in determining such a surface is very high especially in metal cluster systems. In this context, great attention is paid to the computation of DFT-descriptors as indicators of reactivity and stability. This procedure may avoid the detailed study of the energy hypersurface.

The chemical potential (μ) characterizes the escaping tendency of electrons from equilibrium, and it is the Lagrange multiplier associated with the normalization constraint in DFT that has been identified as the negative of the electronegativity ($\mu=-\chi$).[35] On the other hand the hardness (η) can be seen as a resistance to charge transfer and softness ($S=1/\eta$) that has been qualitatively related to the polarizability (α) of the system.[30,36–38]

Although the use of DFT-based reactivity concepts in the field of solid state systems is quite meager, there are a few interesting applications. The hardness concept has been used by Del Pino *et al.*[39] to study segregation of As and Ga in germanium crystallites, and by Mendizabal *et al.*[40] to study the charge capacity of TiS_2 intercalated with Li.

A major focus of attention and discussion in the application of DFT to chemical reactivity is the principle of maximum hardness (PMH) proposed by Pearson.[38] The PMH asserts that molecular systems at equilibrium present the highest value of hardness. Parr and Chattaraj[41] provided a rigorous proof for this principle based on a combination of statistical mechanics and the fluctuation-dissipation theorem, they concluded that the PMH holds under the constraints that the chemical potential and the external potential ($v(r)$) must remain constant upon distortion of molecular structures. However many studies show that the PMH may hold even though μ and $v(r)$ vary during the chemical processes.[42,43]

More recently, on the basis of an inverse relationship

[a)]Electronic mail: atola@puc.cl

 © 2002 American Institute of Physics

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

between hardness and polarizability, Chattaraj *et al.* have proposed a minimum polarizability principle (MPP) (Ref. 44, 45) which states that the natural direction of evolution of any system is toward a state of minimum polarizability. There are many studies confirming the validity of the MPP on different kind of reactions and systems. We have verified the MPP in internal rotations reactions, vibrational dynamics,[45] and proton transfer reactions.[43] More recently, Hohm[46] studied complete fragmentation reactions for a large number of molecules containing atoms of the first and second periods as well as Fe and Os, he found that the most stable isomer has the lowest polarizability in agreements with the MPP.

In the context of the characterization of chemical reactions and reactivity descriptors, the PMH and MPP are widely accepted electronic structures principles, that complement the minimum energy criterion for molecular stablility. With the aim of assessing the use of the aforementioned DFT-concepts and principles in the characterization of solid state systems, we perform in this paper a detailed study of the energetic and electronic properties of Cu_n $(n=1-9)$, with special attention on the performance of the different descriptors to get information on the behavior of copper clusters.

In the next section we present a summary of the theoretical elements we use here; Sec. III contains the computational details; Sec. IV is devoted to the results and discussion; and in Sec. V we draw our conclusions.

II. THEORETICAL BACKGROUND

In DFT, the chemical potential and molecular hardness for the N-electron system with total energy E and external potential $v(r)$ are defined as the following first and second derivatives of the energy with respect to N:[32,35,36]

$$\mu=\left(\frac{\partial E}{\partial N}\right)_{v(\mathbf{r})}=-\chi \tag{1}$$

and

$$\eta=\frac{1}{2}\left(\frac{\partial^2 E}{\partial N^2}\right)_{v(\mathbf{r})}=\frac{1}{2}\left(\frac{\partial \mu}{\partial N}\right)_{v(\mathbf{r})}. \tag{2}$$

In numerical applications, μ and η are calculated through the following approximate versions of Eqs. (1) and (2), based upon the finite difference approximation and the Koopmans theorem,[32]

$$\mu\approx-\tfrac{1}{2}(\mathrm{IP}+\mathrm{EA})\simeq\tfrac{1}{2}(\varepsilon_L+\varepsilon_H) \tag{3}$$

and

$$\eta\approx\tfrac{1}{2}(\mathrm{IP}-\mathrm{EA})\simeq\tfrac{1}{2}(\varepsilon_L-\varepsilon_H). \tag{4}$$

IP is the ionization potential, EA is the electron affinity, ε_H and ε_L are the energies of the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO), respectively. The electrophilicity index is in turn defined as[34]

$$\omega=\frac{\mu^2}{2\eta}. \tag{5}$$

This quantity is a measure of the propensity of a system that acquires electronic charge from the surroundings. Considering clusters or metallic surfaces as binding environments, ω can be useful in analyzing the interaction with small molecules. It is possible to define a local version of ω, by using the additivity property of softness, i.e., $\omega=(\mu^2/2)S=(\mu^2/2)\Sigma_x s_x$, where s_x corresponds to the local softness of the specific site x within the cluster, then $\omega_x=(\mu^2/2)s_x$.

The response of the system is measured by μ and η when N is varied for a fixed $v(r)$. In contrast to this, the polarizability of the system may be used to understand the behavior of the system for changing $v(r)$ at constant N. The mean polarizability is calculated from the polarizability tensor components as

$$\langle\alpha\rangle=\tfrac{1}{3}(\alpha_{XX}+\alpha_{YY}+\alpha_{ZZ}). \tag{6}$$

The static polarizability represents one of the most important observables for the understanding of the electronic properties of clusters, it is proportional to the number of electrons of the systems, and it is very sensitive to the delocalization of valence electrons as well as to the structure and shape of the system.

III. COMPUTATIONAL METHODS

Electronic structure calculations for metals require the treatment of electron correlation, in this context, density functional methods provide a good alternative to post-Hartree–Fock methods. The functionals that we use are the Becke-3 for exchange and Perdew–Wang 91 for correlation (B3PW91),[47,48] and the basis set used for copper atoms is LANL2DZ for effective core potentials.[49] The geometry optimization of copper clusters were performed in redundant internal coordinates as described by Schlegel and co-workers.[50] Vibrational analysis were performed in order to be sure that the structures obtained corresponds to stable clusters. All calculations have been carried out using the GAUSSIAN 98 package.[51]

Chemical potential and hardness were obtained through Eqs. (3) and (4), respectively, using ionization potentials and electron affinities, and the results were compared to $\mu^{(H,L)}$ and $\eta^{(H,L)}$ obtained through the use of the HOMO and LUMO orbital energies. Indeed, we have calculated the energy of cations and anions at the geometry of the neutral systems to determine the vertical $\mathrm{IP}=[E(N-1)-E(N)]$ and $\mathrm{EA}=[E(N)-E(N+1)]$ values from the standard ΔSCF procedure. These values are then used in Eqs. (3) and (4) to obtain what we called $\mu^{(\mathrm{IP,EA})}$ and $\eta^{(\mathrm{IP,EA})}$. Note that in all cases the ground state of neutral and ionic systems correspond to their lowest multiplicity.

The polarizability tensor components were determined analytically as the second derivative of the energy with respect to the cartesian components of the electric field $(\alpha_{ij}=-(\partial^2E/\partial F_i\partial F_j)_{F=0})$. For the calculation of the dipole polarizability, the use of pseudopotentials is well justified because this is primarily a property of the outer shells electrons of the atoms and the core electrons contributions are expected to be negligible.

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

TABLE I. Experimental and theoretical estimations of structural and electronic properties for Cu_2. r_0 = equilibrium distance; ν = vibrational frequency; BE = binding energy; IP = vertical ionization potential; EA = vertical electron affinity. In brackets is reported the % error (expt. − calc.).

Property	B3PW91 (this work)	GGA (this work)	LSDA[a]	LSDA[b]	Experiment
r_0/Å	2.25(−1.35)	2.27(−2.25)	2.18(1.80)	2.21(0.45)	2.22[c]
ν/cm^{-1}	260(1.89)	236(10.94)	292(−10.19)	269(−1.51)	265[c]
BE/eV	1.93(7.21)	2.21(−6.25)	2.72(−30.77)	2.22(−6.73)	2.08[d]
IP/eV	7.8016(1.30)	8.36(−5.77)	8.61(−8.93)	8.73(−10.45)	7.9042±0.0008[e]
EA/eV	0.5880(29.67)	0.93(−11.24)	0.97(−16.03)	1.15(−37.56)	0.836±0.006[f]
				0.96(−14.83)	

[a]Reference 25.
[b]Reference 26.
[c]Reference 27.
[d]Reference 28.
[e]Reference 29.
[f]Reference 18.

IV. RESULTS AND DISCUSSION

A good estimation of ionization potentials and electron affinities is crucial for a reliable estimation of chemical potential and hardness. These properties are quite sensitive to the exchange and correlation functional used in the DFT calculations. In this context, De Proft *et al.*[52] have investigated the performance of many DFT methods in the computation of IP and EA for atomics and simple molecular systems with the results that hybrid functionals gave good estimations for these properties. In light of these data, Perdew–Wang 91 for correlation and Becke-3 for exchange seems to be a reasonable choice for the functionals used in this paper.

First we have tested the validity of the scheme of calculation used here by comparing our results of Cu_2 with previously and recently published theoretical data based on local spin density approximation (LSDA) (Refs. 25, 26) and with the available experimental data.[18,27–29] We have also performed all electron calculations on Cu_2 using the generalized gradients approximation (GGA) with the exchange-correlation functional Perdew–Burke–Ernzerhof[53] and a DZVP2 basis set implemented in the ALLCHEM program.[54] All these results are quoted in Table I.

We first note in Table I that our results of structure, frequency, ionization potential, and binding energy are in good agreement with the experimental and others theoretical data. Only vertical electron affinity is unsatisfactorily estimated. It is well know that electron affinities are considerably more difficult to calculate accurately than ionization energies, basically due to the fact that ionization is an endothermic process while electron attachment can either be endothermic or exothermic. Electron affinities are much smaller in magnitude than ionization energies, and the additional electron tends to make the charge distribution in the anion more diffuse than in neutral and cationic systems. To account for the spatial extent of the outer orbitals, it would be necessary to include more diffuse basis functions in the atomic basis set used in the calculations.[55,56]

In a recent paper Balbuena *et al.*[20] have calculated the adiabatic EA at the same level of theory used by us lowering the error from 30% to 26%. Calaminici *et al.*[22,26] have reported all electron DFT calculations of EA's of copper clusters improving the agreement with experimental data. In Table I, fifth column, are quoted two of their values for EA, the second value was obtained through a scaling procedure that lowered the error to 15%. On the other hand, the all electrons GGA calculations that we have performed on the Cu_2 led to a considerably better value of EA although our original estimation of the other properties were not improved. Taking into account that most properties reported in Table I are in quite good agreement with the experimental data and having reached a fair compromise between the quality of the result and the computational cost, we conclude that the level of calculation chosen is good enough to account for the structural, energetic, and electronic properties of Cu_n ($n=1-9$).

A. Structure and stability of copper clusters

1. The structure of Cu_n ($n=2,9$)

In Fig. 1 are displayed the optimized structures of the ground state of neutral copper clusters from dimer to nonamer. Note that Cu_2 is a 1D structure, Cu_3 to Cu_6 are 2D systems, while Cu_7 to Cu_9 are 3D systems; all these structures are energy minima. In Table II are quoted the values of the average Cu–Cu bond length ($\langle r_{Cu-Cu}\rangle$) and the mean coordination number (CN) with the size of the cluster, CN was calculated as the arithmetic mean of the bond lengths of the first neighbors at each site. The variation of $\langle r_{Cu-Cu}\rangle$ with n shows that structural transitions are occurring when going from 1D to 2D and from 2D to 3D clusters, as illustrated in Fig. 2. As expected, the three-dimentional clusters are those that better approach the experimental distance in the bulk metal (2.556 Å).[57] On the other hand, we note that CN increases linearly with n, and this behavior is indicative of an increase of the electronic delocalization degree and of the binding force.

2. Binding energy

In Table III are displayed the values of total energy for the neutral copper clusters together with the binding energies (BE), or total atomization energies, that have been calculated as

$$BE = nE_{Cu} - E_{Cu_n}. \tag{7}$$

Also included in Table III are the experimentals BE's (Refs. 12, 13) that were determined from a thermodynamical cycle using the results of collision induced dissociation (CID) ex-

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

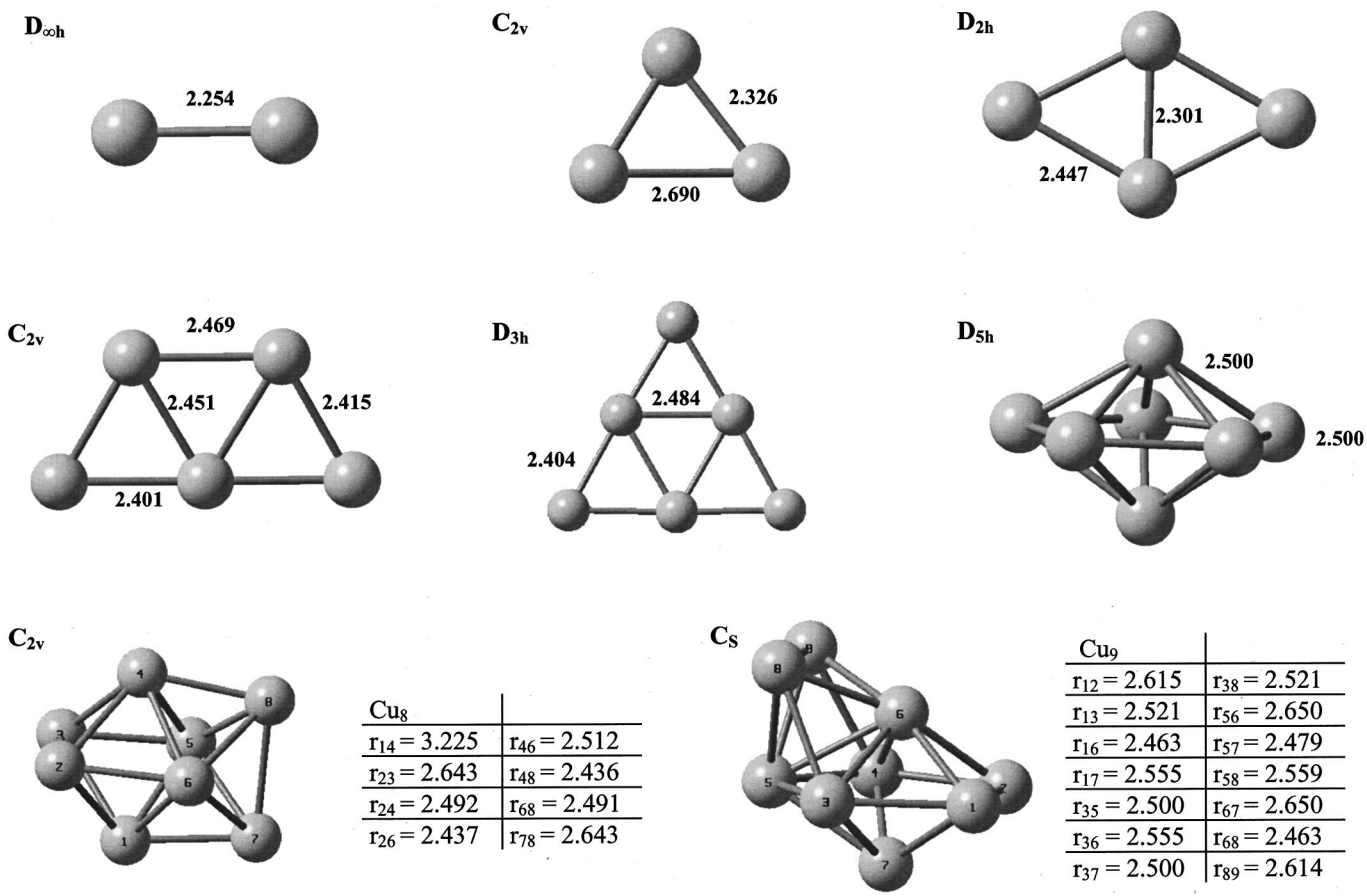


FIG. 1. Ground-state structures of neutral copper clusters, Cu_n ($n=2-9$).

periments of anionic and cationic copper clusters together with electron affinities or detachment energies and ionization potentials of neutral clusters reported in the literature.[10,11,18,29] It is interesting to stress the fact that our BE values lie within the range imposed by the CID experimental values being closer to the CID of anionic clusters. In Fig. 3 we display theoretical and experimental binding energies as a function of the cluster size. We observe that our results follow correctly the quantitative and qualitative trends of the experimental data.

Another property included in Table III is the binding energy per atom (BE/n), that is expected to reach the metal

TABLE II. Average bond length $\langle r_{Cu-Cu}\rangle$ in Å and mean coordination number (CN) of Cu_n clusters.

Cluster	$\langle r_{Cu-Cu}\rangle$	CN
Cu		
Cu_2	2.254	1.0
Cu_3	2.447	2.0
Cu_4	2.418	2.5
Cu_5	2.429	2.8
Cu_6	2.431	3.0
Cu_7	2.500	4.3
Cu_8	2.501	4.5
Cu_9	2.534	5.1
Expt[a]	2.556	12

[a]Reference 57.

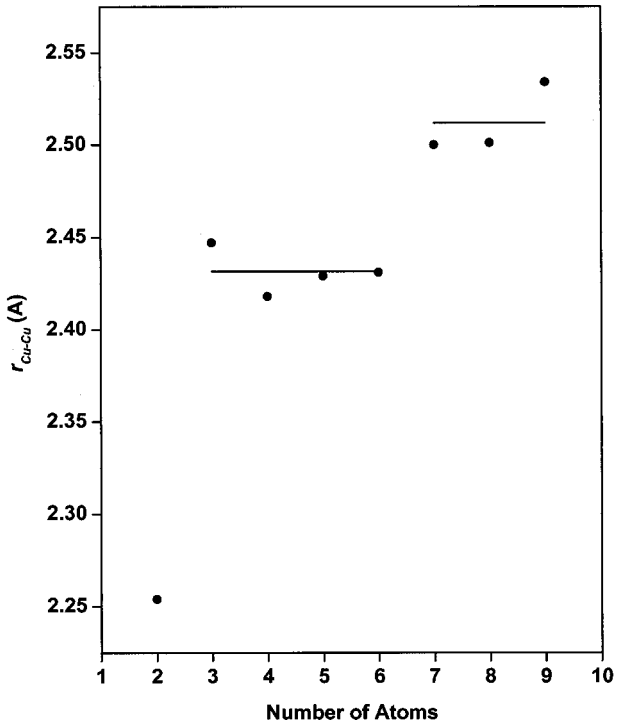


FIG. 2. Dependence of the average Cu–Cu bond length ($\langle r_{Cu-Cu}\rangle$) with the size of the cluster.

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

TABLE III. B3PW91/LANL2DZ energy (a.u.) of the fully optimized structures of copper clusters Cu_n ($n=1-9$) and binding energies BE and BE/n (in eV).

Cu_n	$-E$	BE	BE/n	BE(expt)[a]	BE(expt)[b]
Cu	196.15426				
Cu_2	392.37926	1.9250	0.9625	2.04±0.17	1.81±0.14
Cu_3	588.57325	3.0061	1.0020	3.19±0.26	2.57±0.27
Cu_4	784.80931	5.2320	1.3080	5.91±0.33	4.60±0.81
Cu_5	981.03363	7.1384	1.4277	7.76±0.37	6.19±1.13
Cu_6	1177.27957	9.6332	1.6055	10.32±0.49	7.99±1.37
Cu_7	1373.51538	11.8523	1.6932	12.98±0.66	9.04±1.58
Cu_8	1569.76050	14.3247	1.7906	15.96±0.75	11.20±1.77
Cu_9	1765.97437	15.9468	1.7719		12.22±1.84

[a]CID experiments on the anionic clusters (Ref. 12).
[b]CID experiments on the cationic clusters (Ref. 13).

cohesive energy in the limit of an infinitely large cluster [the experimental cohesive energy of copper is 3.50 eV (Ref. 57)]. It is clear that the clusters studied here are still too small to produce a reasonable estimation of the cohesive energy of copper, however, it can be easily shown that our BE/n values lie within the region defined by the CID experimental data. The agreement observed with the experimental data[12,13] and other theoretical results[22,25,26] is certainly validating the methodology used in this paper.

On the other hand, note that BE and BE/n increase with the size of the cluster, and we observe a maximum value of BE/n in Cu_8. It is well known that the magnitude of BE/n gives information about the strength of chemical bonds in clusters, thus it might help to determine the stability and reactivity of the system. In metallic clusters the increase of this property is due to a high electronic delocalization which is originated by the high coordination of the atoms. From the CN and BE/n values quoted in Tables II and III, we obtain a good linear relation between BE/n and CN: BE/n(CN) $=0.7413+0.2234\cdot$CN ($r=0.94$). Now, introducing the experimental cohesive energy of copper (BE/$n=3.50$ eV) into the above linear equation we estimate the average value of CN to be quite close to CN=12, which corresponds to the experimental coordination number in the bulk. This indicates that the analytic form produced from our results is reliable enough to help rationalize experimental data. We conclude that the level of calculation used throughout this paper is adequate for characterizing binding energies and may provide correct estimations of the DFT-descriptors we are interested in.

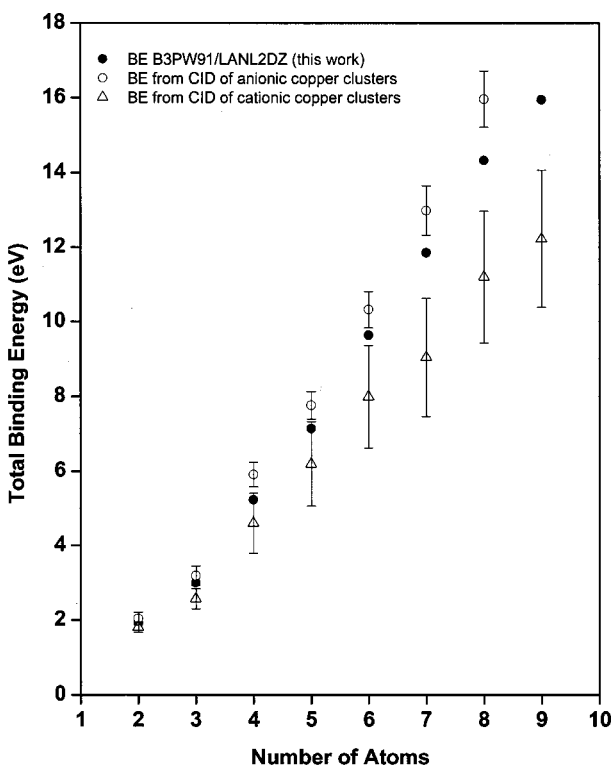


FIG. 3. Experimental CID and theoretical binding energies (BE) as functions of the size of the cluster.

3. Relative stability

The stability of metal clusters is studied through the energy difference,

$$\Delta_2 E(n)=E(n+1)+E(n-1)-2E(n), \tag{8}$$

where n is the number of atoms in the cluster. This quantity is known as the relative binding energy of a cluster with n atoms respect to those with $(n+1)$ and $(n-1)$ atoms. High values of $\Delta_2 E(n)$ indicate that the cluster Cu_n is more stable than its neighbors Cu_{n-1} and Cu_{n+1}. In Fig. 4 we show $\Delta_2 E$

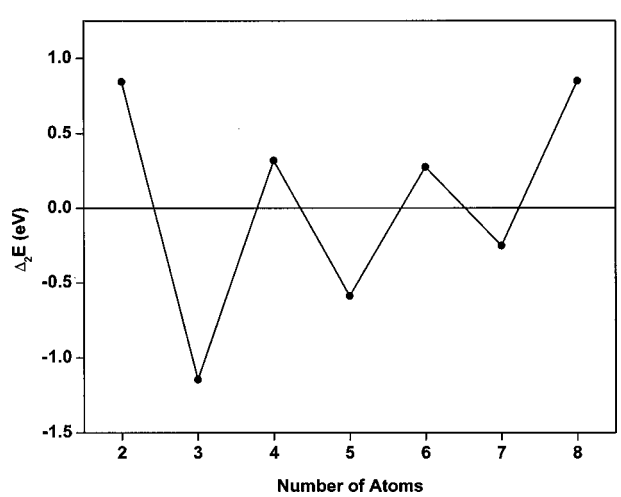


FIG. 4. Second energy difference ($\Delta_2 E$) as function of size of the cluster.

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

TABLE IV. B3PW91/LANL2DZ energies (a.u.) and frontier orbital energies (eV) for the neutral and ionic clusters Cu_n $(n=1-9)$.

Cu_n	$-E$	$-E_{\text{cation}}$	$-E_{\text{anion}}$	$-\varepsilon_H$	$-\varepsilon_L$
Cu	196.15426	195.87159	196.17788	5.1824	2.7522
Cu_2	392.37926	392.09256	392.40087	5.4962	2.1122
Cu_3	588.57325	588.36121	588.60133	3.9756	2.5905
Cu_4	784.80931	784.57161	784.85392	4.6823	2.6711
Cu_5	981.03363	980.80719	981.09078	4.5384	2.9655
Cu_6	1177.27957	1177.02022	1177.31292	5.4246	2.0967
Cu_7	1373.51538	1373.29607	1373.57336	4.4175	2.9802
Cu_8	1569.76050	1569.51666	1569.80145	5.0978	2.3753
Cu_9	1765.97437	1765.77822	1766.00064	3.9304	2.6901

as a function of n, we observe an oscillating behavior and identify clusters with even n as being more stable [$\Delta_2 E(n)$ present local maxima] than the clusters with odd n. This is so because clusters with an even n present closed-shell states.

In Table IV are displayed the values of energy for the N, $(N-1)$, and $(N+1)$ electron systems together with the highest occupied (ε_H) and lowest unoccupied molecular orbital (ε_L) energies. We note that the frontier MO energies present a similar oscillating behavior as observed for $\Delta_2 E$. Clusters with an even n have a large HOMO-LUMO energy gap and therefore are expected to be less reactive than clusters with an odd number of atoms. As already mentioned, the stability exhibited by even clusters is due to their closed-shell configurations that always comes along with an extra stability. It is important to mention that this result is in agreement with the electronic shell jellium model,[58,59] where filled-shells cluster with 2, 8, 18, 20, 40, 58, 92,... valence electrons have increased stability, the mass spectra of cluster distribution shows pronounced intensity in clusters with these number of atoms, the so-called magic numbers.[58,59]

B. Electronic properties

1. Ionization potential

Table V displays our theoretical results together with the available experimental values[10,11,18,29] of vertical ionization potentials and electron affinities of the nine copper clusters studied in this work.

We first note the oscillating behavior of the IP which is due to the change of spin multiplicity of the ground state of this series, and clusters with n even are closed-shell systems, whereas odd-numbered copper clusters are open-shell systems. Therefore even clusters present the higher values of the IP with respect to their neighboring odd systems, because it is more difficult to remove an electron from the doubly occupied HOMO of a closed-shell system than from a single occupied HOMO of an open-shell system. This result is consistent with the variation of HOMO energy along the series (see Table IV).

Our computed ionization potentials show an overall very good quantitative agreement with the experimental results, as can be seen in Fig. 5(a), where a nice linear correlation of experimental and theoretical results is obtained. At this point it is interesting to check the validity of the Koopman's theorem for estimating ionization potentials from the energy of the HOMO. Figure 5(b) shows a fairly good linear relation between the experimental IPs and the HOMO energies. However, the IP's determined using the Koopman's theorem applied to the Kohn–Sham HOMO orbital are systematically underestimating the experimental value (see Tables IV and V), in agreement with recent studies aimed at comparing Hartree–Fock and Kohn–Sham orbitals.[60] It is well known that in a Kohn–Sham calculation the approach of IP by the HOMO energy is a measure of the quality of the functional employed, and that only at the limit of the exact functional, $\text{IP}=-\varepsilon_H$. In this context the satisfactory agreement shown

TABLE V. Vertical ionization potentials and electron affinities for Cu_n. All values are in eV.

Cu_n	IP	IP(expt)	EA	EA(expt)[a]
Cu	7.6919	7.724[b]	0.6427	1.235 ± 0.005
Cu_2	7.8016	7.9042 ± 0.0008[b]	0.5880	0.836 ± 0.006
Cu_3	5.7700	5.80 ± 0.04[c]	0.7641	2.37 ± 0.01
Cu_4	6.4682	7.15 ± 0.75[d]	1.2139	1.45 ± 0.05
Cu_5	6.1618	6.3 ± 0.1[c]	1.5551	1.94 ± 0.05
Cu_6	7.0573	7.15 ± 0.75[d]	0.9075	1.96 ± 0.05
Cu_7	5.9678	6.1 ± 0.05[c]	1.5777	2.16 ± 0.1
Cu_8	6.6353	7.15 ± 0.75[d]	1.1143	1.57 ± 0.05
Cu_9	5.3376	5.35 ± 0.05[c]	0.7149	2.40 ± 0.05

[a]Reference 18.
[b]Reference 29.
[c]Reference 10.
[d]Reference 11.

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

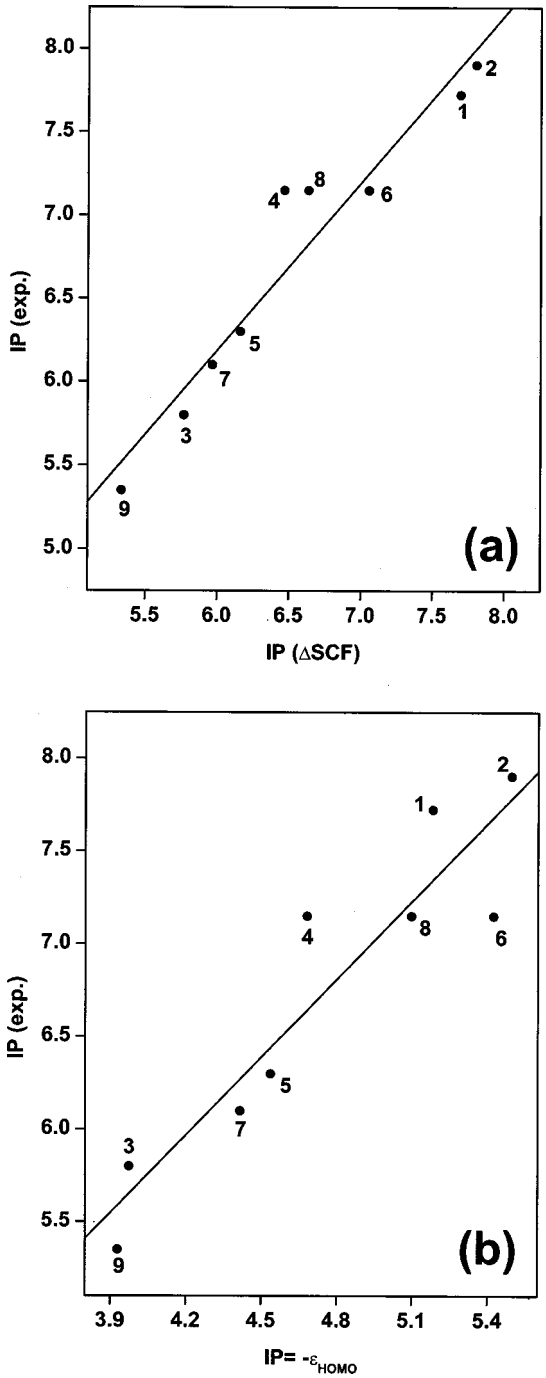


FIG. 5. Comparison of experimental and theoretical ionization potentials along the series Cu_n ($n=1-9$): (a) theoretical IPs from ΔSCF ($r=0.96$); (b) theoretical IPs from HOMO energies ($r=0.93$). All values are in eV.

in Figs. 5 confirm the reliability of our results and the overall adequateness of the employed methodology.

2. Chemical potential, molecular hardness, and electrophilicity

After discussing the structure, the relative stability of the differents clusters and the determination of ionization potentials and electron affinities, we can focus our attention on the characterization and rationalization of copper clusters through the use of DFT-descriptors and electronic structure principles of chemical reactivity. We are interested in whether one can safely introduce these concepts to rationalize the behavior of the clusters.

In Table VI we display experimental and computed values of μ, η, and ω obtained from Eqs. (3), (4), and (5) with experimental and computed values of IP and EA, and using the frontier orbital energies (ε_H and ε_L). We have performed the error analysis for the resulting properties obtained from the experimental data (IP and EA) given in Table V.

Figure 6 shows the behavior of property $P(=\mu,\eta,\omega)$ with n. Calculations using ionization potentials and electron affinities given in Table V give rise to $P^{(\mathrm{IP,EA})}$; the use of frontier orbitals energies given in Table IV led to $P^{(H,L)}$; as already mentioned, the experimental values $[P^{(\mathrm{expt})}]$ were obtained from of the experimental IP and EA values given in Table V. Table VI shows that Cu_4, Cu_6, and Cu_8 present the larger error bars in the experimental data, this is due to the 10% error coming out from their experimental ionization potentials.

The chemical potential is related to charge transfer from a system of high μ to another system with a lower value of the chemical potential. Therefore we expect the clusters with odd n to present maximum values of μ because they are open shell systems and after the transfer of one electron these systems will close their electronic shell, and the resulting systems will be more stable than the original open-shell clusters (see Table IV). In Fig. 6(a) we note that the experimental values of μ follow the expected trend, an oscillating behavior with local maxima in clusters with odd n. In particular Cu_3 and Cu_9 present the highest values of $\mu^{(\mathrm{exp})}$, and these systems are able to transfer an electron to close their electronic shell leading to systems with 2 and 8 valence electrons, respectively. This again is in agreement with the magic number prediction for stability in the jellium model.

Chemical hardness has been established as an electronic quantity that in many cases may be used to characterize the

TABLE VI. Chemical potential, chemical hardness, and electrophilicity index of Cu_n ($n=1-9$). All values are in eV.

Cu_n	$\mu^{(\mathrm{IP,EA})}$	$\mu^{(H,L)}$	$\mu^{(\mathrm{expt})}$	$\eta^{(\mathrm{IP,EA})}$	$\eta^{(H,L)}$	$\eta^{(\mathrm{expt})}$	$\omega^{(\mathrm{IP,EA})}$	$\omega^{(H,L)}$	$\omega^{(\mathrm{expt})}$
Cu	−4.1673	−3.9673	−4.4795±0.0050	3.5246	1.2151	3.2445±0.0050	2.4636	6.4765	3.0923±0.0068
Cu_2	−4.1948	−3.8042	−4.3701±0.0061	3.6068	1.6920	3.5341±0.0061	2.4393	4.2765	2.7019±0.0071
Cu_3	−3.2671	−3.2831	−4.0850±0.0412	2.5030	0.6925	1.7150±0.0412	2.1322	7.7820	4.8651±0.1359
Cu_4	−3.8411	−3.6767	−4.3000±0.7517	2.6272	1.0056	2.8500±0.7517	2.8079	6.7214	3.2439±1.1727
Cu_5	−3.8585	−3.7519	−4.1200±0.1118	2.3034	0.7864	2.1800±0.1118	3.2318	8.9501	3.8932±0.2494
Cu_6	−3.9824	−3.7606	−4.5550±0.7517	3.0749	1.6640	2.5950±0.7517	2.5789	4.2496	3.9977±1.4871
Cu_7	−3.7728	−3.6989	−4.1300±0.1118	2.1951	0.7187	1.9700±0.1118	3.2422	9.5189	4.3292±0.2964
Cu_8	−3.8748	−3.7353	−4.3600±0.7517	2.7605	1.3625	2.7900±0.7517	2.7195	5.1203	3.4067±1.2379
Cu_9	−3.0263	−3.3103	−3.8750±0.0707	2.3114	0.6202	1.4750±0.0707	1.9812	8.8349	5.0900±0.2771

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

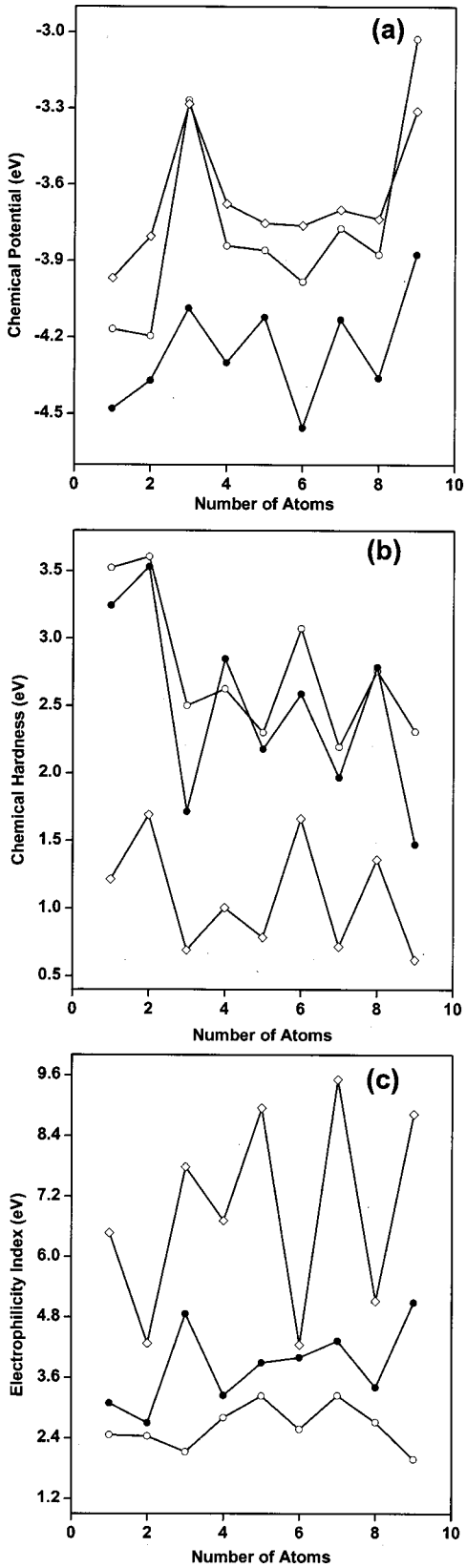


FIG. 6. (a) Chemical potential, (b) chemical hardness, and (c) electrophilicity index as function of size of the cluster. [$P^{(\mathrm{IP,EA})}=-\bigcirc-$; $P^{(\mathrm{exp})}=-\bullet-$; $P^{(H,L)}=-\diamond-$, $P=\mu,\eta,\omega$].

relative stability of molecules and aggregates through the PMH.[41–43] Assuming that the PMH holds in these systems, we expect the hardness to present an oscillanting behavior with local maxima at the clusters with even n, as found for the relative energy in Fig. 4. Figure 6(b) shows that closed-shell even clusters present higher values of hardness than their odd open-shell neighboring clusters. We observe the even–odd oscillating feature similar to that already stressed in Fig. 4. Stable clusters (n even) are harder than their neighbors systems (n odd). These results confirm the statement that the stability of even clusters is determined by the closed electronic shell structure as a manifestation of the PMH.[61,62] Our results are also in agreement with those recently reported by Mineva *et al.*,[63] where the same even–odd alternation rule for hardness was found in sodium clusters.

Finally, it is important to mention that hardness shows quite a stable behavior and qualitative trends of $\eta^{(\mathrm{IP,EA})}$ are quite the same as $\eta^{(H,L)}$, thus validating the use of the two forms Eq. (4). On the other hand we note that in most cases $\eta^{(\mathrm{IP,EA})}$ and $\eta^{(H,L)}$ define the limits where the experimental data lies, and this suggests specific computation strategies to predict upper and lower boundaries for experimental hardness.

The EA's values determined in our calculation undermine the corresponding electrophilicity value, so in the general discussion of this property we discard the $\omega^{(\mathrm{IP,EA})}$ data. The computed electrophilicity indexes $\omega^{(H,L)}$ present a general behavior that is quite close to the one featured by the experimental data as is shown in Fig. 6(c). Along the series we observe an even–odd oscillatory trend, and the maximum electrophilic power is found at the odd clusters because of their already mentioned capacity to close the electronic shell by receiving an electron. Copper clusters with odd n are more willing to accept electronic charge from the environment than do even clusters.

3. Polarizability

We have mentioned that the polarizability can be used to understand the behavior of the system when the external potential is changed at constant N. This property is considered one of the most important observables for understading the electronic properties of clusters. In this context, Moullet *et al.*[64] have reported LDA calculations of the electric dipole polarizability tensor for several isomers of sodium clusters. They have shown how the comparison of the calculated polarizability with the experimental data can be used to identify the specific isomer actually observed in the experiments.

In Table VII we show the average B3PW91/LANL2DZ polarizability calculated using Eq. (6) and the mean polarizability per atom ($\langle\alpha\rangle/n$). Unfortunately, there is no measurement of static electric polarizabilities for these systems to compare with our results. However, polarizability calculations on copper clusters have been carried out by many authors.[23,65,66] Polarizability is very sensitive to the quality of the basis set[67] and it is sometimes recommended to use diffuse functions to describe the distortions of the electron density due to the external electric field. However, sometimes the errors inherent to the theoretical method are in an opposite direction with respect to the errors produced by the basis set. Hence a careful calibration of the basis set can achieve a compensation of errors producing a final result hopefully better than expected.

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

TABLE VII. Static mean polarizablity $\langle\alpha\rangle$ and mean polarizability per atom ($\langle\alpha\rangle/n$) of copper clusters. All values are in a.u.

Cu_n	$\langle\alpha\rangle$	$\langle\alpha\rangle/n$
Cu	52.610	52.610
Cu_2	77.574	38.787
Cu_3	137.148	45.716
Cu_4	153.948	38.487
Cu_5	196.506	39.301
Cu_6	221.212	36.869
Cu_7	242.873	34.696
Cu_8	267.640	33.455
Cu_9	312.970	34.774

First of all let us briefly review the previous calculations. Pou-Amérigo *et al.*[65] have investigated the polarizability of the copper atom using the modified coupled pair functional (MCPF) method and a finite field approach with an external field of 0.01 a.u. They used five differents contracted basis sets ($5s4p3d2f1g$,$6s5p4d3f2g$,$7s6p5d4f3g$, $8s7p6d5f4g$,$21s15p10d6f4g$) which yield to the following polarizabilities of atomic copper: 30.48 a.u., 40.72 a.u., 50.0 a.u., 52.96 a.u., and 53.44 a.u., respectively. It is clear that our calculated value at the B3PW91/LANL2DZ level of theory (52.61 a.u.) is fairly close to the best values of Pou-Amérigo *et al.* Also Sadlej *et al.*[66] have calculated the polarizability of few metals using high-level-correlated-relativistic calculations. The reported value for the copper atom was 46.50 a.u., again in very good agreement with our present result. More recently Calaminici *et al.*[23] have presented a GGA all electrons study of static polarizabilities and polarizability anisotropy calculations of copper clusters with a triple zeta plus valence polarization basis set (TZVP) augmented with seven field-induced polarization (FIP) functions. They obtained $\langle\alpha\rangle=47.02$ a.u. for Cu quite close to Sadlej, Pou-Amérigo, and our own calculations. This satisfactory agreement of our results with all electrons results indicates that the core contributions to $\langle\alpha\rangle$ are quite negligible and reliable polarizability values at a reasonable computational cost can be obtained using DFT combined with pseudopotentials and a double zeta basis set. On the other hand we have check the stability of the polarizability values with respect to the grid quality by performing calculations on Cu and Cu_2 using three differents grids (coarse, fine, and ultrafine in GAUSSIAN 98 program). The results show that the polarizability values are independent of the choice of the grid.

In Fig. 7 we display $\langle\alpha\rangle$ and $\langle\alpha\rangle/n$ with the cluster size. In Fig. 7(a) we note that when going from Cu to Cu_9 the polarizability of the copper clusters increases monotonically showing the expected proportionality with n (or the total electrons number). We note in Table VII that $\langle\alpha\rangle/n$ oscillates when going from Cu to Cu_5, and afterwards it looks like it starts to converge to a value of about 34 a.u. We have fitted these results to an exponential decay of first order, i.e., $\langle\alpha\rangle/n=A+Be^{-Cn}$, and the results are displayed in Fig. 7(b). From this fitting we can predict a value of 31.14 a.u. for $\langle\alpha\rangle/n$ in the copper's bulk.

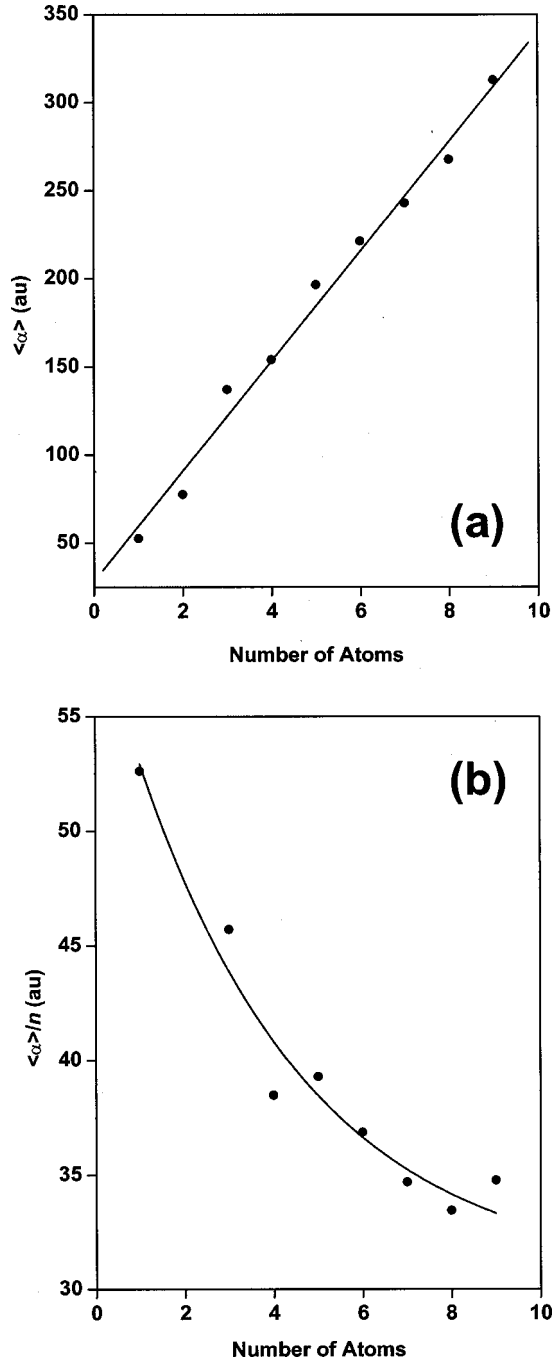


FIG. 7. (a) Calculated polarizability ($r=0.99$) and (b) polarizability per atom ($r=0.96$) as function of the size of the cluster.

C. Chemical reactivity: The PMH and MPP

We have already mentioned that Cu_8 presents the highest value of cohesive energy and the lowest value of polarizability per atom in agreement with the minimum polarizability principle.

To show that consistency between energetic and electronic descriptors has been reached, in Fig. 8 we show a good linear relation between BE/n and $\langle\alpha\rangle/n$ indicating that systems with strong bonds (or high electronic delocalization degree) present minimum values of polarizability per atom, in agreement with the MPP. On the other hand, Cu_8 also

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

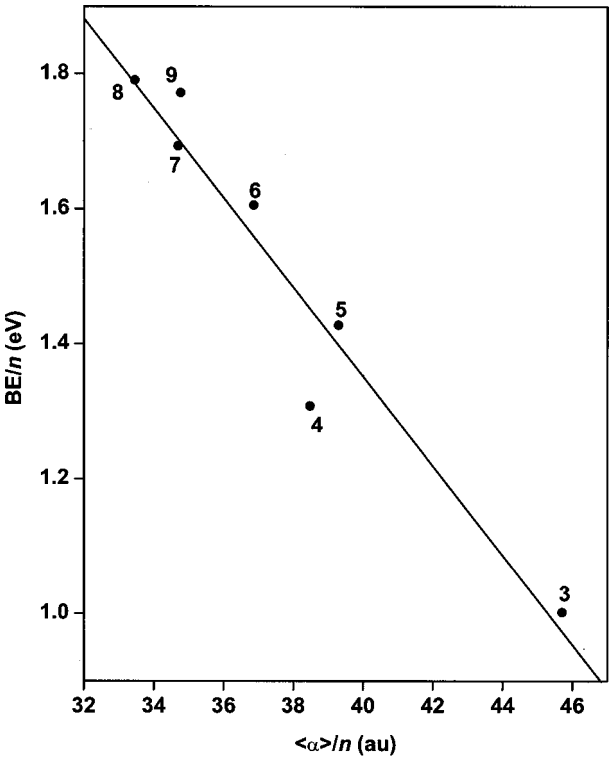


FIG. 8. Correlation between the binding energy per atom and the polarizablity per atom along the series ($n=1-9$; $r=-0.97$).

presents a maximum value of η with respect to its neighbors, in agreement with the PMH.

V. CONCLUSIONS

We have studied small neutral copper clusters at the B3PW91 level of theory combined with pseudopotentials. Our results of geometry, binding energy, and ionization potential showed good agreement with experimental and others theoretical data. The general trend that we have found is that clusters with an even number of copper atoms are more stable (or less reactive) than clusters with an odd number of atoms.

The reactivity descriptors presented in this paper are useful tools in the rationalization of theoretical and experimental data, and this allows us to get new insights on more complex systems involved in heterogeneous catalysis and in cluster's fragmentation reactions. We have found that the more stable is the cluster the less polarizable and the harder it is, in agreement with MPP and PMH, respectively. This shows that the minimum polarizability and maximum hardness principles are operative in the characterization of these kinds of systems.

ACKNOWLEDGMENTS

The authors wish to thank Professor P. Calaminici, A. M. Koster, and A. Vela (CINVESTAV, México) for helpful discussions. This work was supported by Cátedra Presidencial en Ciencias 1998 awarded to ATL and by FONDECYT through Projects Nos. 2000050 and 1020534. P.J. is grateful to Departamento de Posgrado y Postitulo-U. de Chile, Beca PG13/99 and PG20/2000.

[1] M. D. Morse, Chem. Rev. **86**, 1049 (1986).
[2] A. G. Zacarias, M. Castro, J. M. Tour, and J. M. Seminario, J. Phys. Chem. A **103**, 7692 (1999).
[3] J. M. Seminario and J. M. Tour, Int. J. Quantum Chem. **65**, 749 (1997).
[4] G. M. Koretsky and M. B. Knickelbein, J. Chem. Phys. **106**, 9810 (1997).
[5] R. Fournier, J. Chem. Phys. **155**, 2165 (2001).
[6] W. D. Knight, K. Clemenger, W. A. de Heer, and W. A. Saunders, Phys. Rev. Lett. **52**, 2141 (1984).
[7] W. A. Saunders, K. Clemenger, W. A. de Heer, and W. D. Knight, Phys. Rev. B **32**, 1366 (1985).
[8] V. Kreibeg and M. Vollmer, *Optical Properties of Metal Clusters* (Springer, Berlin, 1995).
[9] K. D. Bonin and V. V. Kresin, *Electric Dipole Polarizabilities of Atoms, Molecules, and Clusters* (World Scientific, Singapore, 1997).
[10] M. B. Knickelbein, Chem. Phys. Lett. **192**, 129 (1992).
[11] D. E. Powers, S. G. Hansen, M. E. Geusic, D. L. Michalopoulos, and R. E. Smalley, J. Chem. Phys. **78**, 2866 (1983).
[12] V. A. Spasov, T-H. Lee, and K. M. Ervin, J. Chem. Phys. **112**, 1713 (2000).
[13] O. Ingólfsson, U. Busolt, and K. Sugawara, J. Chem. Phys. **112**, 4613 (2000).
[14] L. Holmgren, H. Gronbeck, M. Andersson, and A. Rosén, Phys. Rev. B **53**, 16644 (1996).
[15] B. J. Winter, E. K. Parks, and S. J. Riley, J. Chem. Phys. **94**, 8618 (1991).
[16] V. Lammers and G. Borstel, Phys. Rev. B **49**, 17360 (1994).
[17] P. Fuentealba and Y. Simón-Manso, Chem. Phys. Lett. **314**, 108 (1999).
[18] J. Ho, K. M. Ervin, and W. C. Lineberger, J. Chem. Phys. **93**, 6987 (1990).
[19] H. Akeby, I. Panas, L. G. M. Pettersson, P. Siegbahn, and U. Wahlgren, J. Phys. Chem. **94**, 5471 (1990).
[20] P. Balbuena, P. Derosa, and J. M. Seminario, J. Phys. Chem. B **103**, 2830 (1999).
[21] L. Padilla-Campos, A. Toro-Labbé, and J. Maruani, Surf. Sci. **384**, 24 (1997); L. Padilla-Campos and A. Toro-Labbé, J. Mol. Struct.: THEOCHEM **390**, 183 (1997); L. Padilla-Campos and A. Toro-Labbé, J. Chem. Phys. **108**, 6458 (1998).
[22] P. Calaminici, A. M. Koster, N. Russo, and D. R. Salahub, J. Chem. Phys. **105**, 9546 (1996); P. Calaminici, K. Jug, and A. M. Koster, *ibid.* **109**, 7756 (1998).
[23] P. Calaminici, A. M. Koster, and A. Vela, J. Chem. Phys. **113**, 2199 (2000).
[24] C. Mossobrio, A. Pasquarello, and A. Dal Corso, J. Chem. Phys. **109**, 6626 (1998).
[25] K. A. Jackson, Phys. Rev. B **47**, 9715 (1993).
[26] K. Jug, B. Zimmermann, P. Calaminici, and A. M. Koster, J. Chem. Phys. **116**, 4497 (2002).
[27] D. G. Leopold, J. Ho, and W. C. Lineberger, J. Chem. Phys. **86**, 1715 (1987).
[28] E. A. Rohlfing and J. J. Valentini, J. Chem. Phys. **84**, 6560 (1986).
[29] A. M. James, G. W. Lemire, and P. R. R. Langridge-Smith, Chem. Phys. Lett. **277**, 503 (1994).
[30] *Structure and Bonding: Chemical Hardness*, edited by K. D. Sen (Springer-Verlag, Berlin, 1993).
[31] P. K. Chattaraj, G. H. Liu, and R. G. Parr, Chem. Phys. Lett. **237**, 171 (1995); P. K. Chattaraj, Proc. Indian Natl. Sci. Acad.-Part A **62**, 513 (1996); R. G. Pearson and W. E. Palke, J. Phys. Chem. **96**, 3283 (1992); T. K. Ghanty and S. K. Ghosh, *ibid.* **97**, 4951 (1993).
[32] R. G. Parr and W. Yang, *Density Functional Theory of Atoms and Molecules* (Oxford University Press, New York, 1989).
[33] R. G. Parr and W. Yang, J. Am. Chem. Soc. **106**, 4049 (1984); W. Yang and W. Mortier, *ibid.* **108**, 5708 (1986).
[34] R. G. Parr, L. von Szentpaly, and S. Liu, J. Am. Chem. Soc. **121**, 1922 (1999).
[35] R. G. Parr, R. A. Donnelly, M. Levy, and W. E. Palke, J. Chem. Phys. **68**, 3801 (1978).
[36] R. G. Parr and R. G. Pearson, J. Am. Chem. Soc. **105**, 7512 (1983).
[37] R. G. Pearson, Inorg. Chem. **23**, 4675 (1984); J. Am. Chem. Soc. **110**, 7684 (1988).
[38] R. G. Pearson, *Chemical Hardness: Applications from Molecules to Solids* (Wiley-VCH, Weinheim, 1997).

Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions

[39] A. Dal Pino, M. Galván, T. A. Arias, and J. D. Joaunopoulos, J. Chem. Phys. **98**, 1606 (1993).
[40] F. Mendizabal, R. Contreras, and A. Aizman, Int. J. Quantum Chem. **56**, 819 (1995).
[41] R. G. Parr and P. K. Chattaraj, J. Am. Chem. Soc. **113**, 1854 (1991).
[42] G. I. Cárdenas-Jirón, J. Lahsen, and A. Toro-Labbé, J. Phys. Chem. **99**, 5325 (1995); G. I. Cárdenas-Jirón and A. Toro-Labbé, *ibid.* **99**, 12730 (1995); S. Gutiérrez-Oliva, J. R. Letelier, and A. Toro-Labbé, Mol. Phys. **96**, 61 (1999); P. Pérez and A. Toro-Labbé, J. Phys. Chem. A **104**, 1557 (2000); M. Torrent-Sucarrat, J. M. Luis, M. Duran, and M. Solà, J. Am. Chem. Soc. **103**, 7951 (2001).
[43] P. Pérez and A. Toro-Labbé, Theor. Chem. Acc. **105**, 422 (2001); P. Jaque and A. Toro-Labbé, J. Phys. Chem. A **104**, 995 (2000); S. Gutiérrez-Oliva, P. Jaque, A. Toro-Labbé, J. Phys. Chem. A **104**, 8955 (2000).
[44] P. K. Chattaraj and S. Sengupta, J. Phys. Chem. **100**, 16126 (1996); P. K. Chattaraj and A. Poddar, J. Phys. Chem. A **102**, 9944 (1998); **103**, 1274 (1999).
[45] P. K. Chattaraj, P. Fuentealba, P. Jaque, and A. Toro-Labbé, J. Phys. Chem. A **103**, 9307 (1999).
[46] U. Hohm, J. Phys. Chem. A **104**, 8418 (2000).
[47] J. P. Perdew and W. R. Wang, Phys. Rev. B **45**, 13244 (1992).
[48] A. D. Becke, J. Chem. Phys. **98**, 5648 (1993).
[49] P. J. Hay and W. R. Wadt, J. Chem. Phys. **82**, 270 (1985); W. R. Wadt and P. J. Hay, *ibid.* **82**, 284 (1985).
[50] C. Peng, P. Y. Ayala, H. B. Schlegel, and M. J. Frisch, J. Comput. Chem. **17**, 49 (1996); C. Peng and H. B. Schlegel, Isr. J. Chem. **33**, 449 (1994).
[51] J. M. Frisch, G. W. Trucks, H. B. Schlegel *et al.* GAUSSIAN 98. Revision A9. Gaussian, Inc., Pittsburgh, PA, 1998.
[52] F. De Proft and P. Geerlings, J. Chem. Phys. **106**, 3270 (1997).
[53] J. P. Perdew, K. Burke, and M. Ernzerhof, Phys. Rev. Lett. **77**, 3865 (1996).
[54] A. M. Koster, M. Krack, M. Leboeuf, and B. Zimmermann, ALLCHEM (Universitat Hannover, Germany, 1998).
[55] A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry* (Dover, New York, 1996).
[56] J-M. André, D. H. Mosley, M-C. André, B. Champagne, M. Clementi, J. G. Fripiat, L. Leherte, L. Pisani, D. P. Vercauteren, and M. Vracko, *Exploring Aspects of Computational Chemistry* (Presses Universitaires De Namur, Namur, 1997).
[57] C. Kittel, *Introduction to Solid-State Physics*, 4th ed. (Wiley, New York, 1971).
[58] M. Brack, Rev. Mod. Phys. **65**, 677 (1993).
[59] W. de Heer, Rev. Mod. Phys. **65**, 611 (1993).
[60] P. Politzer and F. Abu-Awwad, Theor. Chem. Acc. **99**, 83 (1998).
[61] R. G. Parr and Z. Zhou, Acc. Chem. Res. **26**, 256 (1993).
[62] M. K. Harbola, Proc. Natl. Acad. Sci. U.S.A. **89**, 1036 (1992).
[63] T. Mineva, N. Russo, and M. Toscano, Int. J. Quantum Chem. **80**, 105 (2000).
[64] I. Moullet, J. L. Martins, F. Reuse, and J. Buttet, Phys. Rev. Lett. **65**, 476 (1990).
[65] R. Pou-Amérigo, M. Merchán, I. Nebot-Gil, P.-O. Widmark, and B. O. Ross, Theor. Chem. Acc. **92**, 149 (1995).
[66] P. Neogrady, V. Killo, M. Urban, and A. J. Sadlej, Int. J. Quantum Chem. **63**, 557 (1997).
[67] A. J. Sadlej, Collect. Czech. Chem. Commun. **53**, 1995 (1988).

View publication stats
Downloaded 10 Oct 2012 to 200.27.72.254. Redistribution subject to AIP license or copyright; see http://jcp.aip.org/about/rights_and_permissions